

⚡ 30-Day Co-op AI Playbook & Executive

Reference

From Conversational Chatbots to Bounded, Auditable IT Workflows

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Core Thesis: Taking AI to the next level is not about prompting chatbots; it is about building bounded, auditable IT workflows where AI retrieves, analyzes, and prepares work under strict human governance—while keeping the Operational Technology (OT) air gap absolute.

1. The Sacred Boundary: OT vs. IT Critical Isolation

[OPERATIONAL TECHNOLOGY (OT)] [ENTERPRISE IT NETWORK] SCADA • Substations • Breakers CIS/Billing • Ticketing • Docs | |
OPERATIONAL CAPABILITY • Zero control access • Bounded Agentic Pipelines • NERC CIP ESP contained • Local Inference (Ollama/vLLM)
• Relays unidirectional • Mandatory Human-in-the-Loop Gate

Rule 1: AI enhances Enterprise IT productivity; AI never touches the breaker loop. Keeping AI workloads strictly on Enterprise IT subnets prevents expanding the NERC CIP Electronic Security Perimeter (ESP).

2. The 4-Level AI Maturity Ladder

Level	Definition & Mechanics	Utility Cooperative Application
L4: OPERATE FUTURE	Autonomous scheduled pipelines & monitoring	Night-time firewall drift verification, automated log anomaly detection.
L3: BOUNDED EXECUTE TODAY'S HERO	Multi-step agent tools + Human Approval Gates	IT ticket parsing → SOP cross-reference → Staged PowerShell remediation.
L2: GROUND NEXT STEP	Retrieval-Augmented Generation (RAG)	Semantic search across 20 yrs of scanned vendor PDFs (SEL-751) & audio briefings.
L1: ASSIST BASELINE	One-off chat prompts (ChatGPT, base Copilot)	General email drafting, text summarizing (unaware of internal co-op data).

3. The Guardrail Bounded Agent Operating Loop

1. Human Intent & SoT Creation

Sysadmin defines goals & constraints. Orchestrator consults ground-truth Markdown Source of Truth (SoT).

2. Verification Sandbox

Agent lints scripts, tests SQL queries, and checks diffs in isolation before staging any change package.

3. HUMAN APPROVAL GATE

MANDATORY: Execution halts. Sysadmin reviews verified package and clicks 'Approve & Execute'.

4. Production Apply + Git Rollback

Change applied, logged in immutable audit ledger, and backed by instantaneous Git version-control recovery.

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4. The AI Security Contract (6 Governance Guardrails)

1. Data Classification

Tag and filter data prior to ingestion (Public vs. Restricted) to prevent member PII leakage.

2. Least Privilege Tooling

Default read-only tool permissions; write actions strictly scoped to isolated endpoints.

3. Verification Sandboxing

All scripts syntax-checked and linted by secondary critic models prior to human review.

4. Human Approval Gates

High-risk system changes require explicit sysadmin sign-off with clear diff previews.

5. Immutable Audit Logs

Every user prompt, tool call, and script execution recorded in a timestamped compliance ledger.

6. One-Click Git Rollback

Version control (Git) backing every file modification to ensure instant state restoration.

5. On-Premises Hardware Bill of Materials (BoM)

Tier	Hardware Specifications	Model Sizing	CAPEX	Target Utility IT Use Case
Entry Host	Apple Mac Studio (64–128GB Unified)	32B–70B Quantized	\$2.5k–\$4.5k	Private document search, sensitive script drafting, zero cloud leak.
Dedicated Node	NVIDIA DGX Spark / Compact Node	70B Swarms	~\$4.7k	Departmental inference server, continuous log RAG, internal API.
High-Throughput	Dual RTX 4090 / RTX 6000 Ada	70B+ FP16 High-Conc	\$8k–\$18k	High-volume OCR document pipelines, multi-user co-op portal.

*Operating costs: Local inference power draw is ~\$15–\$45/month, compared to \$35–\$95/mo for metered cloud APIs.

6. The 4-Week Action Sprint (Monday Morning Checklist)

WEEK 1: SELECT & CLASSIFY — Pick ONE high-friction, document-heavy IT workflow (e.g. legacy vendor manuals or VPN troubleshooting). Classify data sensitivity (Public vs Internal).

WEEK 2: GROUND & CLEAN — Convert PDFs to clean Markdown; ingest into a grounded tool (Gemini Notebook / Ollama). Verify cited retrieval accuracy and test 15-second audio briefings.

WEEK 3: PILOT BOUNDED LOOP — Deploy a Level 3 agent workflow with a mandatory **Human Approval Gate**. The agent stages PowerShell commands or ticket responses; the admin signs off.

WEEK 4: MEASURE & EVALUATE — Measure technician hours saved, audit error rate, calculate ROI, and present findings to GM/Board before purchasing new hardware.

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